Module 13) >>Python Fundamentals

- ## What is Python ? :

Python is a popular, easy-to-learn programming language used for a wide range of applications,

from web development and data science to automation and artificial intelligence.

- ## Key Features of Python :

1.Simple and Easy to Learn:

Python code looks almost like plain English.

2. High-Level Language:

Use the interpritre And compiler

3. Interpreted Language:

it's line by Line execute the code

4. Multipurpose:

its used in Diffrent platform like :

Web Development

Data Science & AI

Game Development

Desktop Apps

- ## History and Evolution of Python:

Creator: Guido van Rossum:

in 1980 Think for Python Project:

and 1989 Start the python Project

20 feb 1991 lunch the first version of python : First Release: Python 0.9.0

Letest version of Python : 2 oct 2023 : Python 3.12

- **Advantages of Python Over Other Programming Languages:**

python has become one of the most popular progarmmin language due to its simplicity,

versatility, and powerful ecosystem. Here's why it stands out comared to other

language like java, c++, javascript, and Rubby.

1. Easy to Learn & Readable Syntax
2. Versatile & Multipurpose
3. Cross-Platform & Portable
4. Dynamically Typed
5. Faster Development Speed

- **Installing Python and Setting Up a Development Environment:**

1. Installing Python:

> Download Python from the official website:

→ https://www.python.org/downloads/

> Run the installer and check:

> Verify Installation:

python --version

2. Development Environment:

Option 1: Anaconda (Best for Data Science & ML):

installtion steps:

1. Download Anaconda: https://www.anaconda.com/download
2. Run the installer.
3. open Anaconda Navigator.
4. Launch Jupyter Notebook / Spyder for coding.

> Verify Installation:

conda --version

jupyter notebook

Option 2: PyCharm (Best Full-Featured Python IDE):

Installation Steps:

1.Download PyCharm:

https://www.jetbrains.com/pycharm/download/

2. Run the installer.
3. Create a New Project >>Select Python interpreter.
4. Start coding with full IDE features.

Option 3: VS Code:

installtion steps:

1. Download VS code on the Browser.
2. Install the Python extension (by Microsoft).
3. Open a .py file, and VS Code will detect Python.
4. Run Python scripts.

- **Writing and executing your first Python program.**

```
print("Hello, World!")
```

1. Open a terminal/command prompt
2. Navigate to the directory where you saved
3. python (file Name)